# Mahmoud Alyosify

## AI Security & Threat Intelligence Engineer

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## SUMMARY

AI security engineer applying machine learning to **threat intelligence, OSINT and fraud detection**. Trained in AI for Cybersecurity at NTI (420 hours) and applied it at e& Egypt. Built a multi-agent OSINT platform, a threat-intelligence LLM on AWS and a deepfake-audio detector; at Injaz Tech, hardened an LLM assistant against prompt injection with 0 leaks in 8 attack runs. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30).

## SKILLS

**Security:** threat intelligence, OSINT, digital forensics, LLM security (prompt injection, output leaks), guardrails
**Machine learning:** anomaly detection, deepfake detection, classification, LLM fine-tuning (QLoRA)
**Systems:** multi-agent systems, knowledge graphs, RAG, LangChain, LangGraph
**Tools:** Python, PyTorch, TensorFlow, FastAPI, Docker, AWS
**Data:** SQL, MongoDB, Pandas

## EXPERIENCE

**AI Engineer Intern** — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Added prompt-injection filtering, output leak detection and rate limiting; **0 system-prompt leaks** in 8 attack runs.
- Building a RAG assistant that answers employees' questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.

**AI in Cybersecurity Intern** — Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) — Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.
- Completed NTI's **420-hour AI in Cybersecurity** program behind the placement.

**Part-Time Machine Learning Instructor** — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

**Systems Engineer (Military Service)** — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Debugged live transportation software and supported legacy applications to keep downtime to a minimum.

**Online Instructor** — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

## PROJECTS

**HORUS Sentinel — multi-agent threat-intelligence platform** — 2026
*Python, LangGraph, FastAPI, Neo4j, Ollama, Docker*

- Designed a team of AI agents that keep collecting open-source information into a **knowledge graph**; a self-hosted LLM writes evidence-backed reports from it.
- Each agent owns one source; every finding is linked to earlier evidence in the shared graph, so reports trace each claim back to its source.

**Horus-OSINT — fine-tuned LLM on AWS** — 2026
*Python, Llama-3-8B, QLoRA, PySpark, AWS (EC2, S3, EMR), Terraform*

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

- Turned the raw GTD and GDELT records into a training set and fine-tuned the model with **4-bit QLoRA**.

**AudioShield — deepfake audio detection**
*Python, CNN, Streamlit*

- Trained a CNN to detect deepfake (synthetic) speech for voice-fraud and digital-forensics use; live predictions in a Streamlit app.

**MinimalLM — Master's thesis, published on arXiv** 2026
*Python, PyTorch, Hugging Face Transformers, NVIDIA A40*

- MSc thesis: a training-free method that makes open LLMs generate **47.6% fewer tokens** and answer 19–68% faster, with no answer cut off.

Also: SCRAP (satellite collision risk, 48+ hours ahead) · Automated Multimodal Agent (PDF to narrated slides) · SimCLR-Vision-SSL (84.30% top-1, self-supervised).

## EDUCATION

**MSc in Artificial Intelligence** Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

**BSc in Computer & Information Science (Bioinformatics)** Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

## CERTIFICATIONS

- Fortinet Certified Associate in Cybersecurity (2025)
- AI in Cybersecurity (NTI, 420 hours, 2025)
- AWS Certified Cloud Practitioner (Amazon Web Services)
- Generative AI: Boost Your Cybersecurity Career (IBM, 2025)
- IBM AI Developer Professional Certificate (IBM, 2026)
- Deep Learning Specialization (DeepLearning.AI, 2026)

## AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University

## LANGUAGES

Arabic (native) · English (professional working proficiency)